Experimental Design & Scope of Inference

ANSWER KEY

Section 1: Observational Studies vs. Experiments

1 What essential feature distinguishes a randomized experiment from an observational study?

a. **Treatments are actively imposed on experimental units by the researcher ✓**

b. A larger sample size is used

c. Participants are randomly surveyed by phone

d. Data is collected over a longer time period

In an experiment, researchers deliberately apply treatments to experimental units to observe responses. In observational studies, conditions are merely observed without intervention.

2 Researchers observe that people who drink green tea daily have lower rates of cardiovascular disease. Can they conclude that green tea prevents heart disease?

a. **No, observational studies cannot establish cause-and-effect due to potential confounding variables ✓**

b. Yes, because the association is statistically significant

c. Yes, provided at least 10,000 subjects were observed

d. Only if green tea drinkers were matched by age

Causation cannot be inferred from observational studies because lifestyle, diet, or exercise factors may confound the relationship.

3 Only which type of statistical investigation allows researchers to establish cause-and-effect conclusions?

a. **A well-designed randomized comparative experiment ✓**

b. A large-scale stratified observational study

c. A voluntary response poll with $n > 50,000$

d. A census of the population

Only well-controlled experiments with random assignment of treatments can establish causal relationships.

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Section 2: Principles of Experimental Design

4 What is the primary purpose of random assignment in an experiment?

a. **To create roughly equivalent treatment groups and balance out confounding variables ✓**

b. To ensure the sample represents the entire population

c. To eliminate all experimental error

d. To prevent the need for a control group

Random assignment balances the effects of uncontrolled extraneous variables across treatment groups, ensuring differences in response can be attributed to the treatments.

5 What role does a control group serve in an experiment?

a. **Provides a baseline measurement to compare against active treatments ✓**

b. Ensures the experiment has a large enough sample size

c. Guarantees that subjects do not know which treatment they receive

d. Eliminates the need for replication

A control group serves as a baseline comparison to evaluate whether changes in the response variable are truly due to the active treatment.

6 In experimental design, what does 'replication' mean?

a. **Assigning multiple experimental units to each treatment group ✓**

b. Repeating the entire experiment in another country

c. Using identical equipment across all trials

d. Testing subjects multiple times in a row

Replication means applying treatments to sufficiently many experimental units so that treatment effects can be distinguished from chance variation.

Section 3: Blinding, Placebos, and Confounding

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7 What is a double-blind experiment?

a. Neither the subjects nor the individuals evaluating the outcomes know which treatment each subject received ✓

b. Neither the researcher nor the subjects know the purpose of the study

c. The experiment is conducted in a darkened laboratory

d. Both the control group and treatment group receive identical active medications

In a double-blind study, neither the participants nor the evaluators/data recorders know which treatment was assigned, preventing psychological and diagnostic bias.

8 What is the 'placebo effect'?

a. When subjects show improvement simply from the psychological expectation of receiving a treatment ✓

b. When a drug causes adverse side effects in control patients

c. When experimental units drop out of a clinical trial

d. When researchers unconsciously bias results

The placebo effect occurs when patients respond positively to an inactive treatment due to the psychological expectation of healing.

9 What is a confounding variable?

a. A variable related to both the explanatory variable and response variable that obscures whether the treatment caused the outcome ✓

b. A variable that is held constant across all treatment conditions

c. A calculation error during hypothesis testing

d. A categorical variable with more than two levels

A confounding variable is associated with the treatment group assignment and also affects the response, making it impossible to isolate the treatment's effect.

Section 4: Blocking and Scope of Inference

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10 When should an experimenter implement a randomized block design?

a. **When there is an identified characteristic expected to affect the response, so subjects are grouped by that trait before randomization ✓**

b. When the sample size is too small for simple random assignment

c. When double-blinding is completely impossible

d. To avoid having to use a control group

Blocking groups units by a known source of variability (e.g. age, gender) prior to random assignment of treatments, reducing unexplained variation within blocks.

11 A matched pairs design is a special form of:

a. **Block design ✓**

b. Cluster sampling

c. Stratified survey

d. Completely randomized design

Matched pairs is a block design where blocks consist of pairs of closely matched units, or where each subject receives both treatments in random order.

12 If subjects are volunteers (not randomly selected from a population) but are randomly assigned to treatments, what inference can be drawn?

a. **Causation can be inferred, but only for individuals similar to the volunteer participants ✓**

b. Results can be generalized to the entire national population

c. No valid statistical conclusions can be drawn

d. Association only, causation cannot be established

Random assignment allows causal inference; lack of random selection from the population limits generalizability to subjects similar to those who participated.

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